

Database Tables

user

```
id primary key not null

username text not null
hashed_password text not null

firstname text not null
lastname text not null
role enum not null

created_at timestamp not null
achived bool not null default false
```

The user will login with the `username` and `password`.

The `firstname` and `lastname` will be used to display the users name if we need.

The `role` will indicate whether the user will be able to access the elevator controls.

Unique index on `username`.

Roles

- Admin
- Project Member
- Project Supervisor
- ESE Student
- User

can_transaction

```
id primary key not null

sent_by integer not null default 0
FOREIGN KEY sent_by REFERENCES can_node(can_id)

transceived_at timestamp not null
data integer not null

message text null
```

```
current_floor integer not null
last_floor integer not null
```

`sent_by` will reference `can_node` so that all the information about the node can be accessed.
`data` will contain the hex message that was sent.
`message` will contain a text translation of the entire message.

`current_floor` will contain the elevator's floor position when the message was transceived.
`last_floor` will contain the elevator's floor position before it went to the current floor.

elevator_position

```
id primary key not null

current_floor integer null
last_floor integer not null
is_moving bool not null
is_closed bool not null
recorded_at timestamp not null
```

`floor` is the floor number. This will be null if the elevator is moving.

`last_floor` is the floor number of the previous position. This will also display where the elevator came from if it is moving.

`is_moving` will record if the elevator is moving or not.

`is_closed` will record if the elevator's door is open or not.

This table will be added to periodically to record where the elevator is at. Maybe every 10 seconds or so it will be updated. Realistically it wouldn't be updated this often but the elevator is small.

can_node

```
id primary key not null

can_id integer not null
name text not null

created_at timestamp not null
achived bool not null default false
```

Foreign Key so it can be referenced by the `can_transaction` .
Unique Index on `can_id` .